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Turvey's Coordinative Structures Appraised

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ABSTRACT

We present our view on the notion of coordinative structures put forth by Michael Turvey in his ecological-dynamical perspective on perception and action. Starting from two fundamental problems in motor control and coordination, the degrees of freedom problem and the non-univocality of motor commands, we describe how Turvey developed his unique approach to coordinative structures. We present functional specificity and functional integrity as the two main constituent ingredients of coordinative structures in his approach and show that the concepts of lawfulness, self-organisation and coalition follow naturally from these ingredients. Together, these concepts establish a metatheoretical framework that has formed the basis for testable hypotheses at different levels of analysis (e.g. in the movement system or multi-agent system). In the last decades, testing these hypotheses has revealed several properties of coordinative structures at each of these levels with covariation among degrees of freedom being pre-eminent. Challenges of the perspective lie in establishing the relative importance of covariation and understanding the adaptability of coordinative structures through focussing on *de novo* motor learning.

Turvey's coordinative structures appraised

Throughout his career, Michael Turvey developed and promoted an original approach to perception and action, also known as the ecological-dynamical perspective. In this perspective, he combined and expanded the works of Gibson (1966, 1979) and Bernstein (1967) by incorporating dynamical systems theory in the study of motor control and coordination (for an overview of dynamical systems theory see for instance Kugler & Turvey, 1987; Kelso, 1995; Newell, 1996). A central notion in the ecological-dynamical perspective is that of a *synergy*: a temporary assembly of degrees of freedom (DOF) constrained to act as a functional unit. A DOF is a dimension of independent motion in a system, such as an individual muscle that can be activated, an independent direction of mobility in a joint, or an individual's action within a dyad. The notion of a synergy was earlier introduced by Bernstein (1967; Latash, 2021; Latash and

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Turvey, 1996) and has become widespread in the literature over the past five decades (cf. Bruton & O'Dwyer, 2018).¹ Turvey developed his own, unique perspective on synergies that came to have a lasting impact on the field. To distinguish his perspective on a synergy from that of others, in this paper, we will refer to it as *coordinative structure* (c.f. Kugler et al., 1980; c.f. Turvey, 1977, 1990; cf. Turvey et al., 1978). First, we outline how Turvey arrived at his original perspective on motor control and coordination. What were the key ingredients that inspired his thinking? Also, which concepts did he exploit to give substance to the notion of coordinative structures? Second, our goal is to provide a critical appraisal of this notion, explaining its development in the past few decades and its current challenges as we see them.

Ingredients of coordinative structures

In the era in which Turvey started his academic career, the predominant view in the domain of motor control and coordination was that movements were controlled by an executive entity. This entity was supposed to reside in the brain or other parts of the central nervous system of the animal and assumedly controlled the individual DOF in the body. Following his intuition, however, Turvey (see Szokolszky et al., 2023) understood that a theory of motor control and coordination required an account for how action is coupled to perception and vice versa. This led him to argue against the idea of an executive entity, whether it presented itself as a motor program (Schmidt, 1975), an internal model (e.g. Wolpert & Ghahramani, 2000), or as something else. Instead, he promoted a Gibsonian perspective that later formed the foundations of the ecological-dynamical approach (cf. Profeta & Turvey, 2018) and that became a distinctive perspective in the field of motor control and coordination (Feldman, 2023; Rosenbaum, 2023).

In developing his perspective, Turvey started from two key issues in the executive entity account of motor control, as identified by Bernstein (1967; Latash, 2021); Latash and Turvey, 1996). First, the number of DOF in the neuromotor system is just too large for an executive entity to control each individually; in such a scenario the entity has to identify and transmit the appropriate set of motor commands among a plethora of options, which is infeasible in practice. This is commonly known as the DOF problem. Second, the same motor command can lead to different movements, while the same movement can follow from different commands –known as the problem of non-univocality in commands or context-conditioned variability. The core of this problem is that the movement of a limb does not follow from muscular forces alone, but from the sum of the torques created by the muscles, the connective tissue, and the external world (c.f., Bernstein, 1967, Figure 14, p. 20). Because external torques are partly created by events in the environment that are not under the control of the animal, they are only known to the executive entity with a delay (i.e. they are picked up through perceptual systems). Consequently, the motor command can never be accurate to produce the intended movement. Following Bernstein, Turvey thus concluded that a notion of control alternative to that of an executive entity was necessary to explain the coordination of movement.

Turvey also followed Bernstein (1967) in his suggestion that the solution to these two problems lays in the notion of synergy. A synergy, so Bernstein had defined, is a group of muscles that are not anatomically connected but that can be functionally

linked in a unit that is specific for the task at hand (see also Latash, 2021; Latash & Turvey, 1996). A synergy solves the DOF problem because the muscles involved in the synergy are controlled as a unit rather than each individually, thereby compressing the dimensions of the control space. Moreover, in Bernstein's model, synergies coordinate the involved muscles autonomously (Latash, 2021; Latash & Turvey, 1996), without involvement of higher brain centres. They do so by exploiting reflex mechanisms (stretch reflexes and long-loop reflexes) to integrate their activation with fast sensory loops. This enables synergies to respond coherently to external perturbations on a very short timescale. So, in Bernstein's perspective synergies coordinate the DOF at a lower level so that higher levels of control do not need to activate individual DOF.

In advancing his notion of coordinative structures, however, Turvey went far beyond the original ideas on synergies of Bernstein. He exploited his unique position as an ecological psychologist and integrated the insights of Bernstein with Gibson's ideas of perceptual information and affordances (Gibson, 1966, 1979). The two key ingredients that would characterise this integration were functional specificity and functional integrity. Turvey emphasised the continuous influence of perceptual information on the formation of coordinative structures. The environment and the task determine the function that needs to be carried out and, hence, the coordinative structure that needs to be formed. Realising that environmental and task constraints are specified through perceptual information, Turvey concluded that the dynamics of coordinative structures must emerge with reference to the available perceptual information—that is, coordinative structures must be *functionally specified*. In this view, controlling a movement is in fact coordinating action and perception.

Turvey further understood (Kugler & Turvey, 1987; Turvey, 1986, 2007, 2013) that functional specificity implied functional integrity in the coordinative structure. That is, the meaning of a coordinative structure is reflected in the dynamics of the ensemble and not in the activation patterns of the individual muscles. Turvey substantiated this idea by contrasting coordinative structures with the notion of anatomical specificity (Turvey 1986). Anatomical specificity implies that specific anatomical structures produce context-free movements (e.g. reflexes or oscillatory movements). However, so Turvey argued, such anatomical specificity is impossible to reconcile with the observation that different anatomical structures can provide for the same function, while the same structures can produce different functions. Hence, coordinative structures can only be understood as integrated wholes where the role of the parts is determined by the function of the whole. Hence, functional specificity entails functional integrity. Together, the ingredients of functional specificity and functional integrity prompted Turvey to incorporate ideas from thermodynamics, dynamical systems, and self-organisation in the field of motor control and coordination (Kugler et al 1980; Kugler & Turvey 1987; Profeta & Turvey, 2018; Turvey 1990, 2007, 2013). The integration of ecological psychology and dynamical systems theory made Turvey's approach to coordinative structures unique and ground-breaking (see Profeta & Turvey, 2018). But which concepts did he use to shape this perspective?

The concepts underlying coordinative structures

A fundamental concept in Turvey's ecological-dynamical approach is lawfulness. Lawfulness entails interpreting perception and action in terms of regularities among

variables that are coupled through parameters. Lawfulness exists in informational constraints—originating in perceptual flow fields—on actions, but this lawfulness is not the focus here. Goal-directed actions also require lawfully constraining DOF in coordinative structures, and Turvey exploited the concept of self-organisation to explain this process. Self-organisation is prevalent in complex, dynamical systems that are composed of a vast number of components that interact in a nonlinear way, such as movement systems. In such systems, collective behaviour emerges from the interaction between components, while at the same time the collective behaviour constrains the behaviour of the involved components (Haken, 1977; Kelso, 1995; Kugler & Turvey, 1987; Turvey, 2007). This establishes a circular causality through which behavioural patterns with stability properties emerge, also referred to as attractors. By identifying the attractors of a system, it is possible to characterise its current behaviour and its unfolding over time as a function of changing conditions.

Attractors in behaviour follow the laws of complex systems and nonlinear dynamics (cf. Kelso, 2021, 2022). These laws are abstract, in the sense that they are independent from the material properties of the system. Therefore, attractors transcend the boundaries of anatomical, physiological, and physical structures and enable describing the emergent ordered behaviour at different levels of the agent-environment system (c.f., Turvey, 2007; Profeta & Turvey, 2018). With the proper influx of energy, dynamical systems maintain their behaviour (i.e. their order) autonomously, resulting in the systems' behaviour ending up in an attractor state that ensures functional integrity of the coordinative structure (Kugler et al 1980; Kugler & Turvey, 1987; Turvey, 1990).

Given that coordinative structures are temporary assemblages of DOF, they maintain several potential organisations that can be actualised dependent on the situation. Think of a hand that can be organised in a coordinative structure for grasping in one context but for pointing in another. Importantly, the type of coordinative structure that is selected for a task is not instantiated top-down, as in a hierarchical system, but is engendered bottom-up, with all subsystems contributing on par in their interactions. A self-organising process like this requires a specific type of organisation, for which Turvey proposed the concept of a coalition (e.g. Turvey et al., 1978; e.g. Profeta et al., 2021). A coalition, in short, entails that the DOF of both the agent and the environment (i) form a heterarchy—i.e. they are interdependent and on an equal footing with each other (see Turvey, 2007, Figure 1) and (ii) are flexibly organised. Moreover, Turvey claimed that the agent and the environment are dualities, implying complementarity between the two. For instance, an organism can only form coordinative structures if it is complementary to its environment, implying that the dimensions of movement of the organism should match the dimensions of the environment, signifying a symmetric relationship. From this follows mutuality in the control of action as a property of a coalition. Rather than the agent being in control of the environment, subsystems of both the agent and the environment lawfully and reciprocally constrain each other's DOF resulting in task achievement (Profeta et al., 2021; Turvey et al., 1978). This organisational form not only takes the agent-environment system as primary but also emphasises their inherent interrelatedness, ensuring functional specificity and functional integrity of coordinative structures.

Finally, the overview on coordinative structures is not exhaustive, since the insights of Turvey are broader, deeper and more refined than we are able to present here. We hope our exposition inspired others to further take up his insights.

Critical discussion of coordinative structures in an ecological-dynamical perspective

We now turn to developments regarding coordinative structures made within the past four decades. These developments required experimentation, but designing experiments that fully appreciate the metaphysical points of departure of the ecological-dynamic perspective has been challenging. Turvey for one struggled with this point for a long time, as he mentioned open-heartedly in an interview: ‘What was not at all obvious to me was how to deal with it—how to teach Gibson; how to experiment in Gibson’s style. That was not at all obvious to me. I was working all the time towards that. I spent at least ten years trying to figure out how to do this—one decade of my life.’ (Szokolszky et al., 2023, p169).

A challenge for experimenting into coordinative structures is the level of abstraction of the constructs proposed in the ecological-dynamical perspective. This abstraction has motivated researchers to question the falsifiability of coordinative structures as an explanation for motor control and coordination. We postulate that Turvey’s ecological-dynamical perspective should be interpreted as a *metatheory*. Whereas a theory concerns the coherence and structure in a series of empirical findings within a particular scientific area, a metatheory addresses the structure of a set of theories itself (Overton, 2013). The concepts of lawfulness, self-organisation and coalition, as defined in the foregoing section, provide a coherent set of principles that explain what is adequate and meaningful within the ecological-dynamical perspective (c.f., Turvey et al., 1981). Using these concepts as the bedrock, theories about coordinative structures can be devised, tested, and rejected if necessary. The metatheory and its constituent concepts, however, cannot be empirically falsified. As such, these concepts function as a set of axioms in the metatheory that can be debated on theoretical and/or philosophical grounds but not falsified with experiments. Therefore, the ecological-dynamical perspective functions as ‘an engine of discovery’ (Carello & Turvey, 2004, p27) that drives the search for laws of coordination.

To make the step from an abstract metatheory to a concrete theory of coordinative structures, the ecological-dynamical concepts need to be translated to a particular system or level of analysis to define verifiable hypotheses regarding the coordination of DOF. At the level of the movement system, for instance, it has been investigated whether interlimb coordination abides by laws of self-organisation. Rhythmic movements were examined with the goal to reveal the landscape of attractors in the patterns of coordination - think of Kelso’s famous finger wiggling experiments (Haken, et al., 1985; Kelso 2021; 2022). Stable movement patterns were taken as a starting point and studies focussed on which parameters best described these attractors and their stability, which parameters made an attractor switch to another attractor (e.g. from anti-phase to in-phase behaviour), and whether these transitions were characterised by critical fluctuations and hysteresis. The attractor landscape and the transitions were modelled with dynamical equations, which in turn were used to formulate novel and testable

predictions (Beek et al., 1995; Haken et al., 1985; Kelso, 2022). Examples of studies that exploited this approach to examine rhythmic coordination include hand-held pendulum swinging, in which the relative phase between the two pendulums describes the coordinative structure of the emergent, compound pendulum (Turvey et al., 1986). As predicted from dynamical descriptions of such systems, anti-phase swinging was less stable than in-phase swinging, and larger differences between the left and right pendulum (i.e. variations in pendulum length and/or mass) led to larger fluctuations in the relative phase. These findings demonstrated that the coordination of the pendulum swings could be characterised as self-organised (i.e. autonomous) oscillations, in line with other rhythmic activities in humans and in animals (Kelso, 2022). Moreover, similar phenomena were found at the level of multi-agent systems, for instance, in inter-individual leg swinging (Schmidt et al., 1990). To sum, the notion of self-organisation can be used to study and frame coordinative structures within a particular subsystem or at a specific level of analysis and it allows the researcher to empirically validate the structure of coordination while adhering to Turvey's abstract framework.

The enterprise of revealing properties of coordinative structures in rhythmic behaviours (e.g. finger wiggling, pendulum swinging, arm swinging, joint relations in walking or in reaching), exploiting principles of self-organisation, predominantly focused on the emergent behaviour. Therefore, one variable, very often the relative phase between rhythmically moving DOF, was taken as the variable characterising the attractor states in the emergent behaviour. However, focusing on just the emergent behaviour that is reflected in one variable draws attention away from the fact that coordinative structures link DOF (e.g. individual pendulums, joint angles or individual agents) into an ensemble. A full understanding of the notion of coordinative structures therefore requires unpacking the process of coordination on both the level of DOF and the emergent level (Mehta et al. 2025; Tuitert et al., 2020; Valk et al., 2019; Wissing et al, 2020). The Uncontrolled Manifold approach (UCM; Scholz & Schöner, 1999; Schöner, 1995) provides a potent methodology to do this. The UCM analysis takes a performance variable, a parameter of behaviour that is functionally relevant for task achievement, and parses the total variability observed in the individual DOF into variability affecting that performance variable and variability not affecting that variable. When the variability in DOF affecting the performance variable is significantly smaller than that not affecting the performance variable, this evidences that DOF covary to preserve functional integrity of the coordinative structure. A high degree of covariation also implies that a large range of solutions with similar performance outcomes are exploited, indicating flexibility in DOF. As such, the UCM is a very useful tool to translate the metatheoretical concept of a coordinative structure into testable hypotheses regarding motor coordination at different levels of analysis. Before we illustrate this point with examples in two systems, let us explain in more detail the relation between the approaches exploiting the UCM method and Turvey's ecological-dynamical approach.

Originally, the UCM method was developed from a dynamical view on motor coordination (Schöner, 1995). It allowed an analysis of stability in discrete actions, thereby extending the dynamical toolbox (Scholz and Schöner, 1999). Over the years, however, the UCM method has also been adopted by other approaches to motor coordination. Most prominently, (Latash et al. 2007; Latash 2024) developed an

approach to motor control emphasising the flexibility of motor behaviour by applying the UCM method. He suggested to think of the movement system as abundant instead of redundant. Rather than considering the very many DOF as a *problem* for making the system controllable, as the ecological-dynamical approach usually emphasised (Kay, 1988; Riley et al, 2011; Turvey, 1990), he argued that we should consider them as an *opportunity* to adapt to the ever-changing environment. However, we believe that both the redundant and abundant perspectives are compatible with an ecological-dynamical approach. Functional specificity shapes the coordinative structures, limiting the types of behaviour and thereby reducing the control problem. Yet the self-organising principles according to which coordinative structures emerge leave room for flexibility in the organisation of DOF, appreciating abundance. Hence, coordinative structures solve the redundancy problem while exploiting abundance.

In our group, we applied the UCM method to analyse properties of coordinative structures at the level of the movement system. More specifically, we studied goal-directed, discrete arm movements in such as reaching. In tasks like this, the end-effector position of the finger or hand in 3D space is often taken as the performance variable, while the joint angles in the arm are regarded as the system's DOF (e.g. Domkin et al., 2005; Hansen et al., 2015). Using the UCM method, we showed that consistency of the end-effector trajectory co-developed with flexibility of the coordinative structure in reaching movements in middle childhood (Golenia et al., 2018a, 2018b). Also, we found that the coordinative structures that facilitate a goal-directed reaching movement re-organise when the reaching targets are perturbed unexpectedly, and that in most cases this re-organisation preceded the required change in end-effector movement (Wissing et al., 2020). In goal-directed rhythmic pointing, Valk et al. (2019) demonstrated that some joint angles are primary drivers of the end-effector movement whereas other joint angles compensate for variability in the driving joint angles to stabilise end-effector movement. Moreover, the joint angle configurations differed when task constraints differed, indicating that end-effector and joint angles were reciprocally related, suggesting circular causality. Together, these findings show that joint angles covary to maintain the functional integrity of a coordinative structure and that joint angles may take up different roles depending on the coordinative structures' functioning as a whole.

At the level of the multi-agent system, the uncontrolled manifold analysis has been advocated to assess the covariation in motor behaviour *between* interacting individuals (Riley et al., 2011). For instance, Romero et al. (2015) studied the coordinative structure of joint angle configurations across dyads that performed a pointer-to-target aiming task (one participant controls the pointer while the other controls the target). Through a UCM analysis, they demonstrated covariation among the joint angle trajectories of the two participants, which stabilised the pointer-target distance. More recently, Hafkamp et al. (2023), using analyses alternative to UCM, found that the interpersonal structure of coordination in a dyadic ball-and-beam task (two participants control a ball rolling reciprocally on a jointly held beam) was affected by the two participants' task experience. While unexperienced dyads moved the beam sequentially, each taking turns to control the ball, experienced dyads moved the beam concurrently, establishing a cooperative and successful structure of coordination.

All in all, research has revealed one of the main properties of a coordinative structure: degrees of freedom cooperate through covariation to stabilise a task variable. That this property was shown at different levels of analysis, supports our view that the metatheoretical framework of Turvey can be translated in falsifiable hypotheses of coordinative structures. But even though progress has been made, many issues are still outstanding. A pertinent question, in our view, is whether covariation among DOF is the only mechanism underlying the functional integrity of coordinative structures. How does covariation relate to other mechanisms that may preserve a system's integrity? For instance, the functional integrity of the movement system may (partly) result from the contributions of tensegrity principles in connective tissue (Levin, 2006; Levin & Solórzano, 2024; Turvey & Fonseca, 2014). Although, tensegrity principles play a role in the formation of coordinative structures, they do not necessarily result in covariation among the DOF. Also, covariation among members of a multi-individual system during joint action might be mediated or perhaps even overruled by social norms such as the spontaneous division of roles. In other words, it seems likely that the functional integrity of a coordinative structure is ensured by more mechanisms than just covariation. One avenue to advance our understanding of the relative role of covariation in coordinative structures might be to apply complexity measures on UCM variables (Andrade et al., 2024; Grover et al., 2022). Differences in the structure of variability over time might reveal signatures of additional processes at play expanding the perspective of coordinative structures beyond that of just covariation to maintain functional integrity.

We think that a promising route into a more profound understanding of coordinative structures is provided by *de novo* learning. During *de novo* learning, the system reveals its mechanisms for change, which are key to revealing the adaptive functioning of a coordinative structure. Early studies into learning focused on the emergence of new attractors in both simple rhythmic behaviour (Kostrubiec et al., 2012; Leach et al., 2021; Zanone & Kelso, 1992) and complex behaviour like juggling (Huys et al., 2003). Other studies have examined the refining of already acquired coordinative structures in multi-joint tasks (Gray, 2017, 2020; Pacheco & Newell, 2018a, 2018b; Ranganathan & Newell, 2013). More recently, attention has turned to the acquisition of multi-joint coordinative structures in novel tasks (Mehta et al., 2025; c.f. Lee et al., 2018; Lee & Ranganathan, 2019). In these studies, *de novo* learning has been framed as a search in a multi-dimensional DOF space (Bongers, 2021; Pacheco et al., 2019; Ranganathan et al., 2014; Sternad, 2018). Accordingly, the coordinative structure has been conceptualised as a solution space in the DOF space and analyses focus on exploratory processes to arrive at the solution space during practice. Using this rationale, Mehta et al. (2025) demonstrated that a larger part of the DOF space is explored with random practice than blocked practice, in a task where participants had to learn a new mapping between joint angles in the arm and movement of a virtual paddle. We are developing a program around search, exploration and solution spaces to characterise the acquisition of novel multi-joint coordinative structures from an ecological-dynamical perspective.

In this program we also focus on how the use of perceptual information co-evolves with the formation of coordinative structures. We test whether specifying information is picked up before the structure in the coordination has actualised, establishing the coming

into existence of functional specificity. Moreover, we investigate whether covariation reduces linearly or nonlinearly during de novo learning, improving understanding of development of functional integrity. Finally, we believe that de novo learning, and developmental studies for that matter, might create possibilities to empirically address the concept of coalition, which has hardly received attention in the literature. If a coordinative structure for a task might under certain experimental conditions be too difficult to learn, the agent and the environment might be conceived as non-symmetric. This might be a potential direction of empirical research to link the notions of learning and coordinative structure to one of the founding concepts of the ecological-dynamical approach.

To conclude, we interpret the ecological-dynamical perspective advocated by Turvey as a metatheory that provides the ingredients and concepts required to develop theories of temporary and stable structures of coordination in which DOF are functionally organised. The notion of a coordinative structure has been used to form testable hypotheses using the UCM approach at various levels of analysis and this has revealed several key characteristics of motor control and coordination. We are, however, just at the beginning of understanding how DOF are organised into these functional units. Future focus should be on establishing the relative pertinence of covariation and on explaining how novel coordinative structures are formed.

Note

- 1 For instance, the notion of synergy is used to describe relatively fixed modules in the spinal cord that produce a given proportional muscle activation among a group of muscles, which are called muscle synergies (cf Tresch et al., 1999, d'Avella et al., 2003, Bizzi et al., 2008), whereas Latash and colleagues use the notion of synergy to describe the cooperation of DOF to stabilise a performance variable (Latash, 2024; Latash et al 2007).

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